4

SUPPLY, LAW OF SUPPLY AND ELASTICITY OF SUPPLY

QUICK REVISION

1. Supply and Stock: Supply is the quantity of a commodity that a producer is willing to sell in the market in a given period of time at a given price. On the other hand, stock implies the total quantity of the commodity that can be brought into the market by him at a short-notice.

2 . Factors Affecting Supply: The following are the factors, in general, on which the supply of a commodity depends:

(i) Price of the commodity (supply and price of the commodity are directly related. As price increases, supply also increases and vice-versa).

(Ii) Price of related commodities (price of other goods and supply are inversely related. As price of other goods increases, supply decreases and vice-versa).

(iii) Goals of the firm.

(iv) Prices of factors of production used in making the commodity (price of inputs and supply are inversely related. As the price of inputs increases, supply decreases and vice-versa).

(v) State of technology (if there is technological advancement then cost of production falls, profits increases, therefore supply increases).

(vi) Number of producers.

(vii) Future expectations regarding change in price.

(viii) Taxes and subsidies (As excise tax rate increases, supply decreases, hence they are inversely related.)

(ix) Natural factors.

(x) Means of transportation and communication.

3.Supply Function: Supply function expresses the functional relationship between the supply of a commodity and its different determinants, such as price of commodity, price of related goods, price of factors of production, goals of the firm's technology, etc.

$$Sx = f(Px, Pp\ T, GT\&S, G.....)$$

4. Law of Supply: Law of supply states that supply of commodity, other things being equal, varies directly with its price.

5.Supply Schedule: It is a table which shows various quantities of commodity which the producers are willing to produce and sell at various prices during a given period of time.

6. Supply Curve is the Graphic Presentation of Supply Schedule: It is of two types:
(i) Individual supply curve.
(ii) Market supply curve.

7. Reasons For Operation of Law: The Supply curve has positive slope because of the following reasons:
(i) In view of higher profits (in the state of rising prices) producers increase their production and in the state of falling prices, supply decreases because of low profit margin.
(ii) Entry of new firms in a state of rising prices and exit of firms when prices fall.

8. Exceptions to the Law: The law of supply does not apply under the following circumstances:
(i) Vertical straight line parallel to Y-axis, in case of perishable goods and antiques.
(ii) Backward bending supply, in case of supply of laborers.

9. Change in Quantity Supplied and Change in Supply: A change in quantity supplied refers to a movement along the same supply curve. In the movement no new supply curve is drawn whereas change in supply means shift of the supply curve due to change in factors than product price.

10. Extension and Contraction of Supply: Expansion or extension of supply refers to increase in quantity supplied due to rise in price of the goods. Contraction of supply refers to decrease in quantity supplied due to fall in the price of the goods.

11. Increase and Decrease in Supply

Increase in Supply: In a shift of the supply curve, a new supply curve is drawn. When the supply of a commodity increases due to favorable changes in factors other than price of the product's own price, it is called increase in supply. In such a case, there is a rightward shift in the supply curve which is caused by the following factors:
(i) Improvements in technology.
(ii) Fall in the prices of factors of production.
(iii) Changes in the goals of the firms.
(iv) Increase in the number of firms.

Decrease in Supply: When there is a decrease in supply, the supply curve shifts to the leftward. When supply of a commodity falls due to unfavorable changes in factors other than the commodity's own price, it is called decrease in supply. These unfavorable causes may include:
(i) Obsolete technique of production.
(ii) Increase in the price of related goods.
(iii) Increase in the cost of production.
(iv) Decrease in the number of firms.

12. Meaning of Elasticity of Supply: Elasticity of supply refers to the responsiveness of quantity supplied of a commodity to changes in its own price. The formula for measuring elasticity of supply is:

$$Es = \frac{Percentage\ change\ in\ quantity\ supplied}{Percentage\ change\ in\ price}$$

13. Kinds of Price Elasticity of Supply: There can be five different situations of elasticity of supply:

(i) Perfectly Inelastic Supply: Which is indicated by a straight line supply curve parallel toY-axis. This is the case of zero price elasticity of supply.

(ii) Less than Unit (E,<1): Here, the percentage change in quantity supplied is less than the percentage change in price.

(iii) Unit Elastic Supply: Here, percentage change in quantity supplied is greater than percentage change in price.

(iv) Elastic Supply: It illustrates the case when percentage change in quantity supplied is greater than percentage change in price.

(v) Perfectly Elastic Supply: In this case, supply may increase or decrease to any extent without any change in price.

14. Factors Influencing Elasticity of Supply: The following factors affect the elasticity of supply:

(i) Nature of the commodity.

(ii) Time period.

(iii) Availability of resources and facilities.

Graphic Measurement of Elasticity of Supply: Under this method, elasticity of supply is measured at a point on a supply curve. If the supply curve of a commodity passes through the origin it is known as unitary elastic supply. If the supply curve of a commodity cuts the vertical axis, it will be highly elastic supply. If supply curve of a commodity cuts horizontal axis, it will represent a less elastic supply. The formula used in this method is:

$$Es = \frac{Horizontal\ line\ segment}{Total\ quantity}$$

(i) If any straight line supply curve originates from the point of origin, elasticity of supply will be equal to one.

(ii) Elasticity of supply will be greater than one, if straight line supply curve originates from Y-axis.

(iii) If any straight line supply curve meets X-axis to the right of point of origin, elasticity of supply will be less than one.

1 MARK QUESTIONS

A. FILL IN THE BLANKS

1. Total quantity of a commodity available with the seller in the market at a given time is called _____.
2. Part of stock which the seller is willing to offer at a particular price is _____.
3. Advanced technology leads to _____ in supply.
4. More the price of inputs _____ is the supply.
5. The indirect tax on a commodity under new system rises from 15% to 18%, the supply curve will _____.
6. Increase in price leads to _____ of the supply curve.
7. Decrease in price leads to _____ of the supply curve.
8. Change in supply is due to _____ .
9. Change in quantity supplied is due to _____ .
10. Change in quantity supplied is also _____ along the same curve.
11. When quantity supplied does not react to change in price, supply is said to be _____ .
12. Supply curve, parallel to X-axis depicts _____ supply.
13. Straight line supply curve from the point of origin depicts _____ supply.
14. When percentage change in price is less than percentage change in quantity supplied, elasticity of supply is _____ .
15. When % change in price is more than % change in quantity supplied, elasticity of supply is_____ .
16. Other things being equal, direct relation between price and supply is known as _____ .
17. When change in supply is affected by the price alone keeping other things constant, it is known as _____ .
18. The supply curve for perishable goods is _____ .
19. Supply curve for labourers is _____ .

20. When there is reduction in indirect taxes, supply curve shifts _____ .
21. Perfectly _____ supply curve is parallel to X-axis.
22. For perishable goods like milk, elasticity of supply is _____ .
23. Functional relationship between supply and its determinants is known as _____ .
24. Supply follows _____ concept.
25. Supply curve is always _____ sloping.

ANSWERS

1. Stock	2. Supply	3. Increase	4.Less
5. Shift to the left	6. Upward movement	7. Downward movement	8. Change in other factors of supply keeping price of the commodity constant
9. Change in price of the Commodity	10. Movement	11. Perfectly inelastic	12. Perfectly elastic
13. Unitary elastic	14.Greater than one	15. Less than one	16. Law of supply
17. Change in quantity supplied	18.Vertical straight line parallel to Y-axis	19. Backward bending	20.To the right
21. Elastic	22. Zero	23. Supply function	24. Flow

B. TRUE OR FALSE

1. Contraction of supply occurs due to change in factors other than price of the commodity.
Ans. **[False]** Contraction of supply occurs due to fall in price of the given commodity, assuming no change in other factors.

2. Supply is always unitary elastic for all supply curves starting from the origin.
Ans. **[True]** Any supply curve passing through the origin will have unitary elastic supply.

3. In case of zero elastic supply, supply curve is a horizontal straight line.
Ans. **[False]** As in case of a zero elastic supply, supply curve is a vertical straight line parallel to Y-axis.

4. A cost saving technology shifts the supply curve of a commodity to the left.
Ans. **[False]** Cost saving technology will raise the profit margin and will shift the supply curves of the commodity to the right.

5. An increase in excise tax leads to fall in supply of the given commodity, without any change in its market price.
Ans. **[True]** It happens because increase in excise tax, increase the cost of production and reduces the profit margin.

6. A supply curve cutting X-axis has highly elastic supply.
Ans. **[False]** The supply curve cutting X-axis has less elastic supply.

7. Individual supply curve are steeper as compared to market supply curve.
Ans. **[True]** It happens with a change in price, the proportionate change in market supply is more than the proportionate change in individual supplies.

8. If the proportionate change in the quantity supplied of a commodity is less than the proportionate change in price, price elasticity is greater than one.
Ans. **[False]** In such case, price elasticity is less than unity.

9. The supply curve of a goods shifts to the right when prices of other goods rise.
Ans. **[False]** When price of other goods rise, it becomes more profitable to produce them in place of the given goods, so supply curve will shift to left.

10. When price of a factor input falls, supply of the goods will also fall.
Ans. **[False]** Because with fall in price of input, cost of production falls and profit increases. Consequently supply will rise.

11. Law of supply explains quantitative relationship between price and supply.
Ans. **[False]** Law of supply explains qualitative relationship between price and supply.

12. Elasticity of supply explains qualitative relationship between price and supply of the commodity.
Ans. **[False]** There is quantitative relationship between price and quantity supplied of the commodity.

C. MULTIPLE CHOICE QUESTIONS

1. What do we call a quantity which a seller is ready to supply at a given time and at a given price?
(a) Supply (b) Elasticity of supply
(c) Demand (d) Elasticity of demand

2. Other things being equal relation between supply and price is known as:
(a) demand function (b) supply function
(c) (a) or (b) (d) none of these

3. Supply during very short period would be:
(a) perfectly elastic (b) perfectly inelastic
(c) elastic (d) none of these

4. Other things being equal, positive relationship between supply and price is known as:
(a) law of demand (b) law of supply
(c) elasticity of supply (d) none of these

5. In which of following goods, law of supply does not apply?
(a) Perishable goods (b) Medicines
(c) Fruits (d) All of these

6. When change in supply is affected by price alone, it is known as:
(a) movement along demand curve (b) movement along supply curve
(c) elasticity of supply (d) none of these

7. When supply increases due to improvement in technology,
(a) extension in supply (b) increase in supply
(c) contraction in supply (d) decrease in supply

8. When price falls and supply of the product also falls, it
(a) decrease in supply
(b) contraction in supply
(c) extension in supply
(d) any of these

9. Which of the following is the cause of rightward shift of supply?
(a) Improvement in technique of production
(b) Fall in factor price
(c) Rise in price of related goods
(d) All of these

10. Responsiveness of change in supply due to change in price is known as:
(a) elasticity of demand
(b) increase in supply
(c) elasticity of supply
(d) none of these

11. Why is elasticity of supply always positive?
(a) Direct relation between supply and price
(b) Inverse relation between demand and price
(c) Both of these
(d) None of these

12. When supply curve is parallel to Y-axis, elasticity
(a)0
(b) ∞
(c)Es>1
(d) Es<1

13. When supply curve is parallel to X-axis, elasticity
(a)0
(b) Es = 1
(c)Es = ∞
(d) Es > 1

14. When supply curve starts from origin, E, would be:
(a) Es = 1
(b)Es = ∞
(c)Es = 0
(d)Es =< 1

15. When supply curve starts from X-axis, E, would be:
(a) E = 1
(b)Es < 1
(c)Es = ∞
(d)Es = 0

16. When supply curve starts from Y-axis, E, would be:
(a) E>1
(c)Es = ∞
(c)Es = ∞
(d) None of these

17. When there is reduction in indirect tax, supply curve shifts towards:
(a) left
(b) right
(c) any of these
(d) none of these

18. What is the formula of measuring Es ?
(a) $\frac{Percentage\ change\ in\ supply\ of\ commodify}{Percentage\ change\ in\ price}$
(b) $\frac{Percentage\ change\ in\ price}{Percentage\ change\ in\ supply}$
(c) any of these
(d) none of these

19. When supply rises by 50% with an increase in price of 25%, elasticity of supply would be:
(a) 0.5
(b) 2
(c) 3
(d) 0

20. What is relation between stock and supply?
(a) Supply can be equal to stock.
(b) Supply can be less than stock.
(c) Supply can never be exceed stock.
(d) All are correct.

21. Elasticity of supply for a goods is 2. When its price rises by 10%, what would be rise in its supply?
(a) 20%
(b) 5%
(c) 15%
(d) none of these

22. When supply of a goods rises by 40%, its Es=2. What would be a rise in its price?
(a) 20%
(b) 80%
(c) 10%
(d) none of these

23. When price of a goods rises by 20% and supply rise by 10%, E, would be:
(a) elasticity
(b) less elasticity
(c) more elasticity
(d) none of these

24. Supply curve is always:
(a) slope upward left to right
(b) parallel to X-axis
(c) parallel toY-axis
(d) None of these

25. Which of the following is a reason for decrease in supply?
(a) Fall in factor price
(b) Fall in price of related goods
(c) Rise in factor price
(d) None of these

26. What is elasticity of supply for perishable goods?
(a) Less elasticity
(b) Greater elasticity
(c) Perfectly elastic
(d) None of these

27. If E, is unitary, when price of a goods rises by 15% what would be rise in supply?
(a) 8%
(b) 12%
(c) 10%
(d) 15%

ANSWERS

1. (a)	2. (b)	3. (b)	4. (b)	5. (a)	6. (b)	7. (b)	8. (b)	9. (d)	10. (c)
11. (a)	12. (a)	13. (c)	14. (a)	15. (b)	16. (a)	17. (b)	18. (a)	19. (b)	20. (d)
21. (a)	22. (a)	23. (b)	24. (a)	25. (c)	26. (a)	27. (d)			

D. ASSERTION-REASON TYPE

Read the following statements-Assertion (A) and Reason (R), and select the correct alternative in each case:

(a) (A) is true, but (R) is false.
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true and (R) is the correct explanation of (A).
(d) Both (Al and (R) are true but (R) is not the correct explanation of (A).

1. Assertion (A) : Expansion is an upward movement on the same supply curve.
Reason (R) : Keeping all other things constant with increase in the price, quantity supplied increases.

2. Assertion (A) : With the change in price if supply does not change is known as perfectly inelastic supply.
Reason (R) : It happens in case of highly perishable commodities which cannot be stored for a long period of time.

3. Assertion (A) : Contraction in supply leads to a downward movement along the same supply curve.
Reason (R) : Downward movement along the same supply curve occurs due to an increase in price of the commodity, other factors remaining constant.

4. Assertion (A) : Law of supply is a qualitative statement.
Reason (R) : Law of supply indicates the magnitude of change in the quantity supplied.

5. Assertion (A) : In the case of perfectly inelastic supply, the supply curve is a vertical straight line supply curve.
Reason (R) : Supply does not change with change in price in case of E, = 0.

ANSWERS

1. (c)	2. (c)	3. (d)	4. (d)	5. (b)

2 MARK QUESTIONS

1. What do you mean by supply?
Ans. Supply of any commodity means the amount offered for sale by different firms at any particular price during any particular time period.

2. Mention any two factors that determine the supply of a commodity.
Ans. (1) The price of the commodity and (ii) the input prices, determine the supply of a commodity.

3. What is a supply function?
Ans. The supply function shows the interdependence between the supply of a commodity and the factors determining the supply.

$$Sx = f(Px , Pr, Tr, GT\&S, G,.....)$$

4. What do you mean by 'ceteris paribus' supply function?
Ans. The 'ceteris paribus' supply function shows the interdependence between the supply of a commodity and the price of that commodity assuming all other factors determining the supply to remain unchanged.

5. What do you mean by an individual supply schedule?
Ans. The individual supply schedule refers to a table showing different quantities of a commodity supplied at different possible prices by any producer during any particular time period.

6. What is a supply curve?
Ans. The graphical presentation of a supply schedule results in a supply curve.

7. How can you determine the market supply schedule?
Ans. The market supply schedule of a commodity (or the market supply curve) can be determined through the horizontal summation of individual supply schedules of different firms.

8. Define market supply.
Ans. Market supply of a commodity refers to the supply of that commodity by all the firms in the market at a given price during any particular time period.

9. State the law of supply.
Ans. According to the law of supply, the quantity supplied of a commodity increase with the increase in the price of this commodity and the quantity supplied decrease with a fall in its price.

10. What do you mean by an 'extension' in supply?
Ans. An increase in quantity supplied due to an increase in the price of the commodity indicates an 'extension' in supply.

11. What is meaning of a 'contraction' in supply?
Ans. A fall in quantity supplied due to a fall in the price of the commodity indicates a 'contraction' in supply.

12. What do you mean by an 'increase' in supply?
Ans. If the supply of a commodity increases at a given price level, it means an increase in supply.

13. What do you mean by a 'decrease' in supply?
Ans. If the supply of a commodity decreases at a given price level, it means a 'decrease' in supply.

14. What causes an upward movement along a supply curve?
Ans. An increase in quantity supplied due to an increase in price causes an upward movement along a supply curve.

15. What causes a downward movement along a supply curve?
Ans. A decrease in quantity supplied due to a fall in price causes a downward movement along a supply curve.

16. What do you mean by price elasticity of supply?
Ans. The degree of responsiveness of quantity supplied to changes in its own price, is called the price elasticity of supply.

$$Es = \frac{Percentage\ change\ in\ quantity\ supplied}{Percentage\ change\ in\ price}$$

17. What is the meaning of relatively elastic supply?
Ans. When the percentage change in quantity supplied becomes more than the percentage change in its own price, then the supply is considered to be relatively elastic.

18. When does the supply become relatively inelastic?

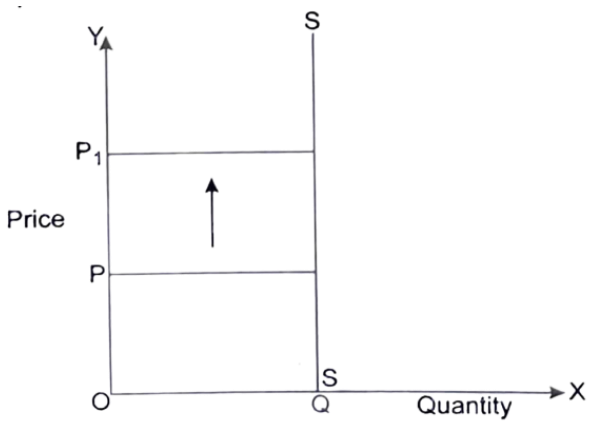
Ans. When the percentage change in quantity supplied is less than the percentage change in its own price, then the supply is considered to be relatively inelastic.

19. What are the two methods of measuring the price elasticity of supply?

Ans. (i) The percentage method and (ii) The geometric method, are used to measure the price elasticity of supply.

20. What will be the shape of the supply curve of a commodity when price elasticity of supply is completely inelastic?

Ans. The supply curve of that commodity will be vertical:



21. State the law of supply.

Ans. The law of supply states, if all other factors, except the price of the commodity, influencing the supply of a commodity remain unchanged, then supply of the commodity increases with an increase in its price and decreases with a decrease in its price.

3-4 MARK QUESTIONS

1. Explain the concept of supply with the help of an example.

Ans. The supply of any commodity means the quantity of a commodity that a firm offers for sale at a particular price.

Price per Unit (₹)	Quantity Supplied(Unit)
5	10
8	17
10	20
17	22

Here, we see that the firm is ready to supply 10 units of the commodity (say, per week) at a price of ₹ 5 per unit; 17 units at a price of ₹ 8 per unit; and so on.

2. Mention any two reasons for the rightward shift of a supply curve.

Ans. The supply curve of any commodity slopes upward from left to right. We can indicate the following two reasons for such upward slope:

(i) **Profit Motive of a Firm**: When the average price of a commodity is higher than the minimum expected price of the firm, it earns some extra profit. Hence, it gets an incentive to supply more at higher prices.

(ii) **An Increasing-cost Industry**: In such an industry, new firm would be willing to supply more only at higher prices to cover additional costs. So, in the long-run, the supply curve of such an industry will be upward sloping.

3. Explain any two reasons for the leftward shift of the supply curve.

Ans. The leftward shift of the supply curve means a decrease in supply at each price level. We can state the following two reasons for such leftward shift:

(i) **Increase in Input Price**: Given the price of a commodity, if there is an increase in input prices then average cost of production will rise. So, a firm would supply less of the commodity than before at a given price.

(ii) **Imposition of Excise Duty by the Government**: If the Government imposes an excise duty on the value of output produced by a firm, its average cost of production will rise. So, at a given product price, the firm is willing to supply less than before.

4. Give any four assumptions of law of supply.

Ans. Main assumption of law of supply is cetris paribus, i.e., other things remaining the same. Other assumptions are:

(i) Price of related goods should remain same.

(ii) Goal of the firm should not change.

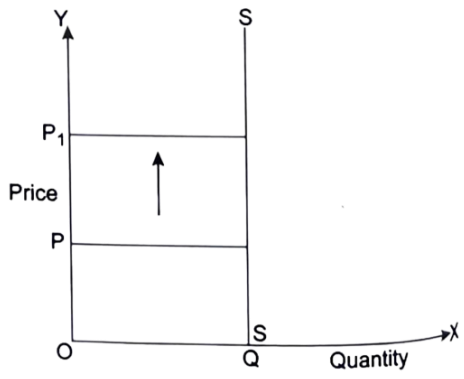
(iii) Input prices should remain same.

(iv) Taxation policies should remain same.

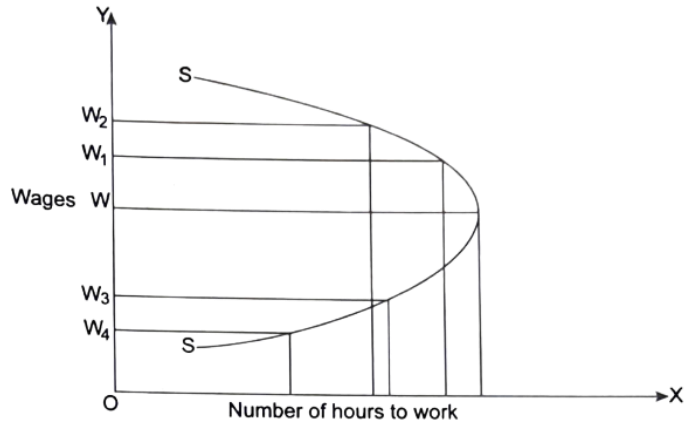
5. Explain two exceptions to the law of supply.

Ans. There are two exceptions to law of supply:

(i) **A Vertical Straight Line Supply Curve, is equal to Y-axis**: It happens in the case of highly perishable commodity, supply for which cannot be increased or decreased due to their highly perishable nature. For example, fish, milk and milk products, leafy vegetables, etc.



(ii) **Backward Sloping Supply Curve**: This type of supply curve might occur in the case of labourers in terms of no. of hours worked. In this case, to some extent, supply of labourer increases with increase in wages but after a certain point supply of labour starts decreasing even if the wages are increasing. This shows after a certain wage-level labourers like to enjoy more leisure than to do work. Thus, this peculiar nature of them gives a backward bending supply curve which is an exception to the law of supply.



6. What do you mean by supply function? Explain.

Ans. Supply Function: Supply function is the functional relationship between quantity supplied of a commodity and its determinants.

$$Sx = f(Px, Pr, T, GT\&S, G.,...)$$

Supply function can be of two types:

(i) **Individual Supply Function**: It refers to the functional relationship between supply of a commodity by an individual producer of a commodity and the factors affecting it. It is expressed as: Sx = f(Px, Pr, St, T, G,.....)

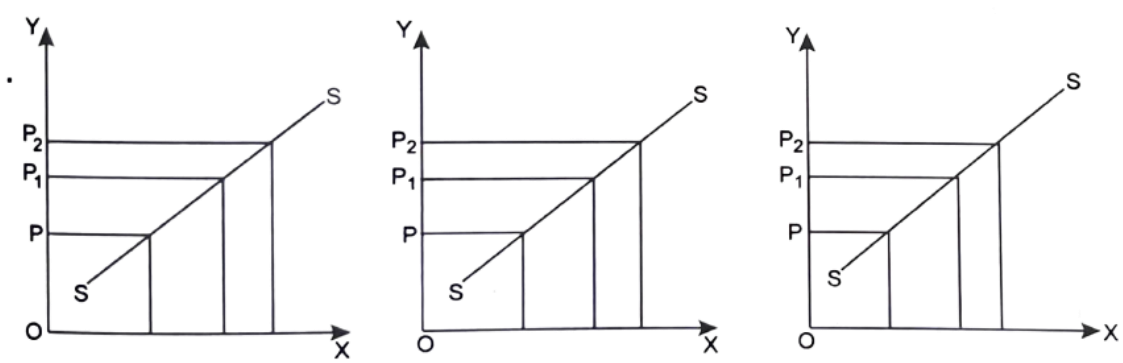
Sx = supply of X commodity
Px = price of X commodity
Pr = price of related goods
St = state of technology
G = goals of the firm
T = level of technology

(ii) Market Supply Function: It refers to the functional relationship between market supply and factors affecting the market supply of a commodity. It can be expressed as:

Sx = f(Px, Pr, Pf , St, T, G, N, F., M,.....)

7. Derive a market supply curve with the help of individual supply curves.

Ans.



8. What is a supply schedule? Explain it with an example.

Ans. The supply schedule indicates different quantities of any commodity offered for sale by any firm or an industry, at different possible price. Thus, we get an individual supply schedule at the firm level and the aggregate supply schedule at the industry (consisting of many firms) level. This is shown below:

	Quantity offered for supply (per month)	
Price (₹ per unit)	**By a firm (units)**	**By an industry (units)**
10	30	2000
12	35	2250
15	42	3050
18	50	4225
20	65	5580

6-8 MARK QUESTIONS

1. Explain any four determinants of supply.

Ans. (i) **Price of the Commodity**: Price and the quantity of the commodity are in direct relationship between each other because the main aim of the producer is to maximize the profit that is why he sells more at higher prices and lesser at low prices.

(ii) **Price of Related Goods**: Supply of commodity also depends upon price of related goods, e.g., substitute goods. Producers always have a tendency to, shift over to substitute goods in case their prices are higher than the price of the commodity sold by the producer.

(iii) **Input Price/Factor Price/ Cost of Production**: Input price is known as the price paid for all types of factor of production. In case of high cost of production with given price, profit margin reduces and the supply of the commodity also reduces. On the other hand, with the decrease in input prices at the given price of the commodity, profit margin increases and the quantity supplied also increases.

(iv) State of Technology: With a better technology efficiency of the producer increases and the output also increases which increases the quantity supplied of the commodity. On the other hand, if there is outdated and inefficient technology used in production, the output decreases and supply will also decrease.

2. Explain law of supply with the help of schedule and curve.

Ans. It states other things being equal supply of a commodity changes directly with the change in its price. Main assumption of law of supply is cetris paribus, i.e., other things remaining the same.

Other assumption are:

(i) Price of related goods should remain same.

(ii) Goal of the firm should not change.

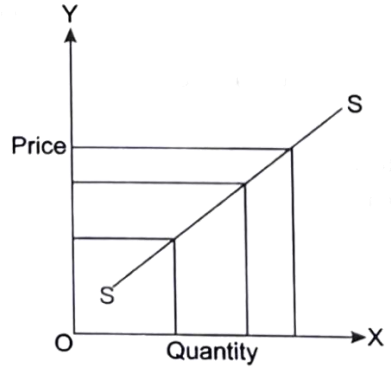
(iii) Input prices should remain same.

(iv) Taxation policies should remain same.

Supply Schedule: It is a table which shows various quantity of commodity which the producers are willing to produce and sell at various prices during a given period of time.

Price	Supply
2	5
4	10
6	15
8	20
10	25

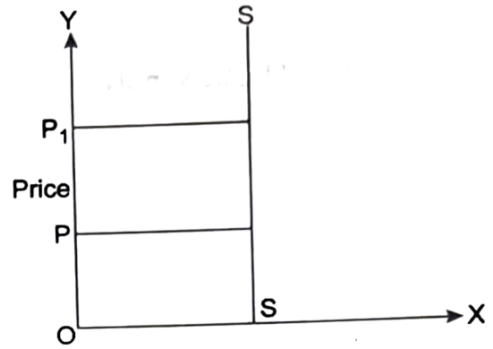
Supply curve in the graphic presentation of supply schedule is:



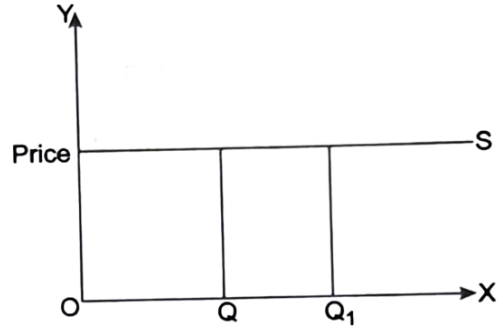
3. Explain all five degrees of elasticity of supply.

Ans. Degree of price elasticity of supply:

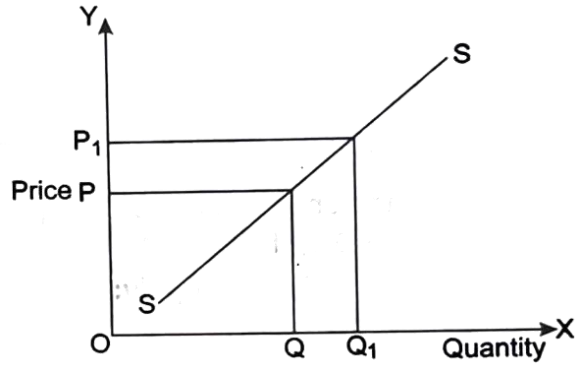
(i) **Perfectly Inelastic**: In this case, change in price does not effect the quantity supplied at all. Supply curve in this case will be a vertical straight line is equal to Y-axis. The elasticity of supply will be zero. It happens in the case of highly perishable goods which cannot be stored for a long period of time.



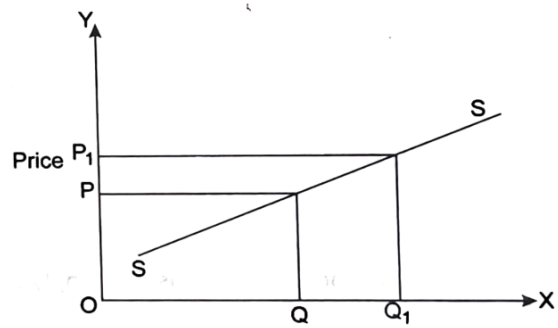
(ii) **Perfectly Elastic**: When quantity supplied changes without any change in the price or little decrease in the price will bring quantity supplied to zero is known as perfectly elastic supply. Supply curve in this case will be a horizontal straight lines equal to X-axis and elasticity of supply = ∞.



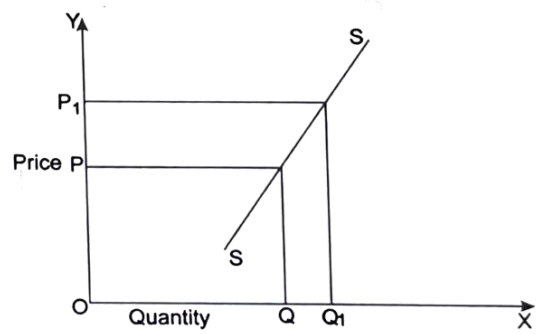
(iii) **Unitary Elastic Supply**: When % change in quantity supplied is equal to the percentage change in price is known as unitary elastic supply. Supply curve in this case will be upward sloping starting from the origin and the elasticity of supply will be 1 (Es= 1).



(iv) **More than Unitary Elastic Or Elastic Supply**: When percentage change in quantity supplied is greater than percentage change in price is known as elastic supply or > unitary elastic supply. Supply curve in this case will be upward sloping starting from Y-axis and Es>1.



(v) **Inelastic Or Less than Unitary Elastic Supply**: When percentage change in quantity, supplied is less than percentage change in price is known as inelastic or less than unitary elastic supply. Curve (in this case) will be upward sloping originating from X-axis Es <1.

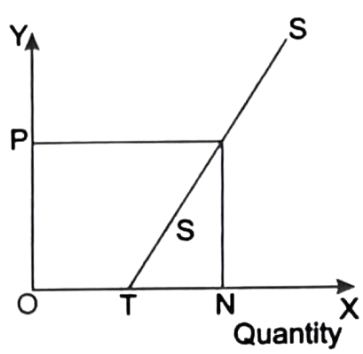


4. Explain point method to measure elasticity of supply.

Ans. Graphic Method: Under this method, elasticity of supply is measured at a point on a supply curve. If supply curve of a commodity passes through the origin, it is known as unitary elastic supply. If supply curve of a commodity cuts vertical axis, it will be highly elastic supply. If supply curve of a commodity cuts horizontal axis, it will represent a less elastic supply. The formula used in this method is:

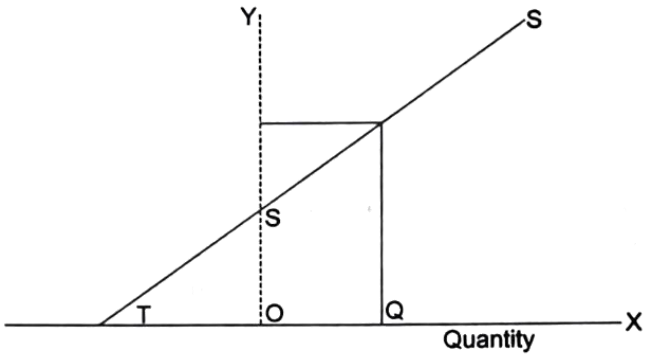
$$Es = \frac{Horizontal\ line\ segment}{Total\ quantity}$$

(i) **Less Elastic Supply:**



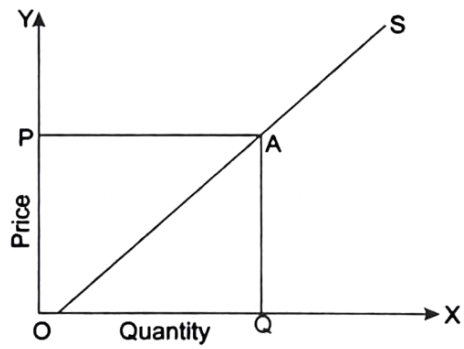
In the above graph SS is the supply curve which cuts OX line at T. Elasticity of supply at point P will be TN/ON. Since, TN<ON, thus Es < 1.

(ii) **Highly Elastic Supply:**



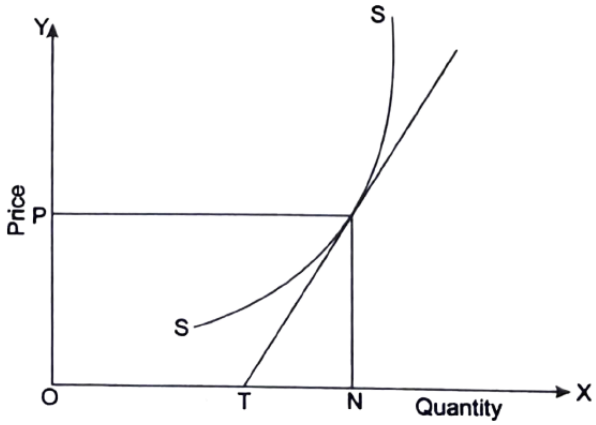
In the above graph SS is supply curve which cuts Y-axis and rests at T. Elasticity of supply at point P will be TQ/OQ, since TQ>OQ, thus, Es,>1.

(iii) **Unitary Elastic Supply:**



In the above graph SS is the supply curve which passes through the origin. Elasticity of supply at P will be TQ/OQ, since TQ=OQ, thus E, = 1.

In case of a Non-Linear Supply Curve:



QUESTION BASED ON DIFFERENTIATION

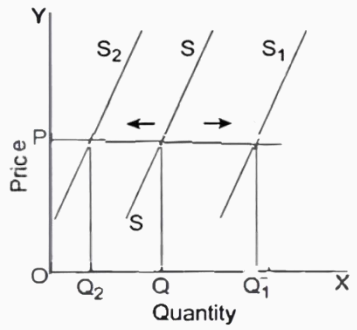
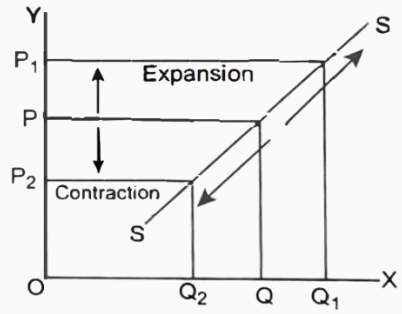
1. Differentiate between cardinal utility and ordinal utility.

Ans.

S.No.	Basis of Differentiate	Stock	Supply
1.	Meaning	Refers to the total amount of goods which producers and sellers are ready do offer for sell at a particular point of time.	Means quantity of commodity that are offered for sale in the market in given period of time at the given price.
2.	Dependence	Stock of the commodity mainly depends on: (i) production of commodity (ii) procurement price (iii) storage and transport cost	Supply of a commodity depends mainly on the market price of the commodity.
3.	Concept	It has a stock concept, i.e., stock refers to amount of a commodity at a particular point of time.	It is a flow concept, i.e., it refers to the amount of a commodity during a period of time.
4.	Commodities	In case of highly perishable commodity, stock and supply would almost the same since these items cannot be stoked for a long period.	In case of durable commodity, supply consists only a part of total stock.
5.	Objective	The stock of any commodity helps in checking fluctuations of market price.	Enables the firm to earn sales revenues.

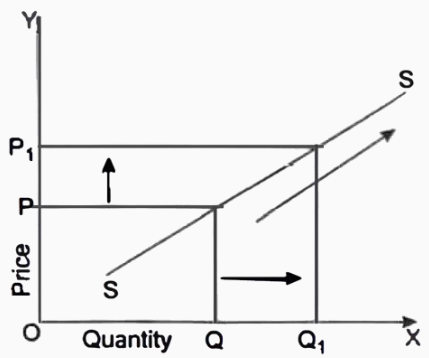
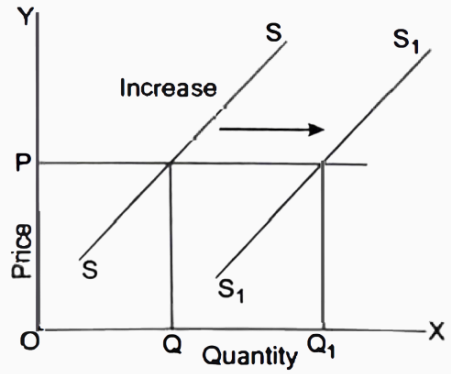
2. Distinguish between change in supply and change in quantity supplied.

Ans.

S.No.	Basis	Change in Supply	Change in Quantity Supplied
1.	Meaning	Here, the supply of a commodity increases or decreases at a given price due to change in other determinants of it.	Here, the quantity supplied increases due to an increase in its price and decreases due to a decrease in its price keeping all other things constant.
2.	Direction of the movement	It causes either a rightward or a leftward shift of the supply curve.	It causes either an upward or a downward movement along the same supply curve.
3.	Types of movements	Increase-Decrease.	Expansion-Contraction.
4.	Graph		

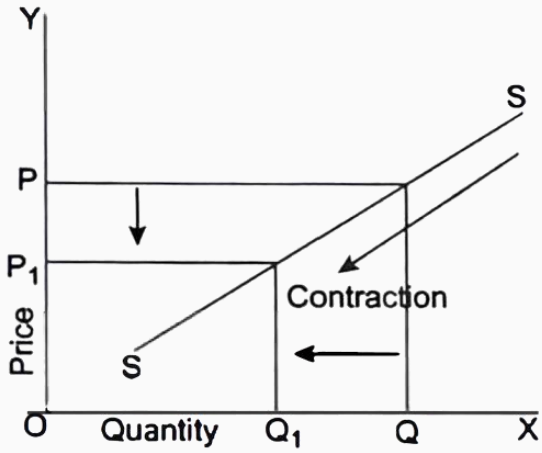
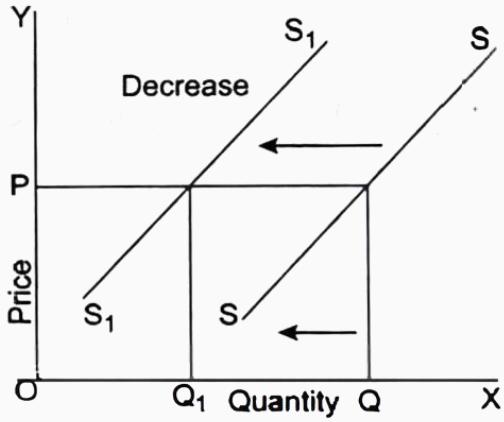
3. State two differences between extension in supply and increase in supply.

Ans.

S.No.	Basis	Extension in Supply	Increase in Supply
1.	Meaning	It shows an increase in quantity supplied due to an increase in price, keeping all other things constant.	It shows a rise in the supply of commodity given the price of the said commodity, due to the changes in other determinants of supply.
2.	Direction of the movement	It implies an upward movement along any given supply curve.	It implies a rightward shift of the supply curve.
3.	Graph		

4. Give two differences between contraction in supply and decrease in supply.

Ans.

S.No.	Basis	Contraction in Supply	Decrease in Supply
1.	Meaning	It shows a decrease in quantity supplied due to a decrease in price of the commodity keeping all other things constant.	It implies a decrease in quantity supplied at a given price due to change in other determinants of supply.
2.	Direction of the movement	It shows a downward movement along the same supply curve.	It implies a leftward shift in the supply curve.
3.	Graph		

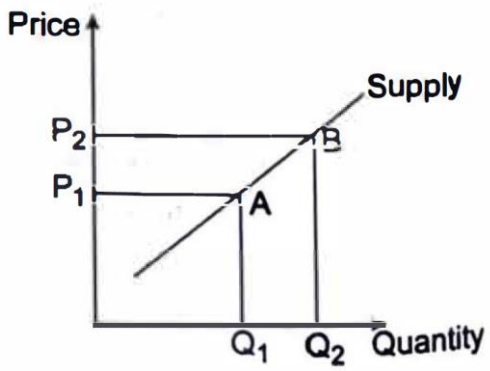
CASE BASED QUESTIONS

CASE 1

Law of supply states positive relationship between price and quantity supplied. It only tells us about the qualitative relation between price and quantity supplied. It explains the effect of change in price on the quantity supplied but not the effect of change in supply on the price. Supply gets affected by many factors and the main factor is price of the commodity.

Answer the following questions:

1. State the law of supply.
2. Write any two assumptions of the law of supply.
3. Show graphically the relation between price and quantity supplied.



Ans. 1. Keeping all other things constant quantity supplied of a commodity increases with increase in price and decreases with decrease in supply.

2. Assumptions are:

(i) Price of related goods remains constant.

(ii) Goals of the firm remains same.

3. Graphical relationship between price and quantity supplied.

CASE 2

Raman is working as a domestic helper in a family and getting 5,000 in a month. He has his parents dependen on him along with his wife and one child. The income earned by him is not sufficient to run his family. To increase the earning of job in a hotel and his income increased to 25,000. Now he asked his parents to stay at home and take rest. He is doing the family his wife also started working even his mother and father also doing some casual work. After few years he got a some extra work in the hotel and earning good amount of money. Now he asked his wife also to stay at home and take care of the whole household things.

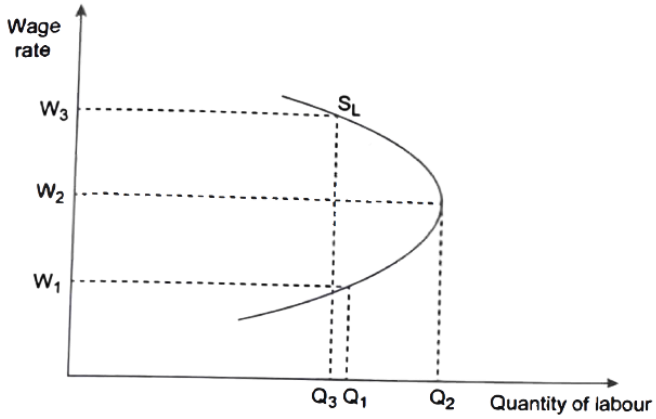
Answer the following questions:

1. Draw the supply curve on the basis of the given case.

2. Is law of supply followed in the given case?

3. Give the reason for the shape of the supply curve based on the given case.

Ans. 1.



2. In the above graph till W_2 law of supply is followed but beyond that it is not.

3. For the above graph the reason is that in the beginning labourers prefer work over leisure but after some time they prefer leisure over work.

CASE 3

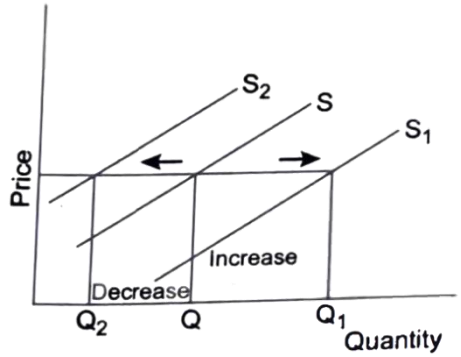
In recent times Indian government has increased the price of petroleum products in the open market. Petroleum products are used as input for various kind of productions. It is used for transporting goods from one place to another also. When the price of petroleum products is high, the cost of production will be increasing and it will affect the supply of the goods also. Supply gets affected by various factors and input price is one of them. It has an inverse relation with supply.

Answer the following questions:

1. What happens to the supply curve when input prices increase?
2. Draw the graph to show the change in supply.
3. Explain two other factors affecting supply of a commodity.

Ans. 1. When input prices increase then due to increase in the cost of production the profit margin decreases and supply curve shifts to the left.

2.



3. (i) State of technology
   (ii) Government policy

QUESTION WITH HIGH DIFFICULTY LEVEL

1. What will be the impact of a rise in sales tax on the supply curve of a commodity?

Ans. The supply curve will shift upward because the seller would then charge higher price for supplying the same quantity.

2. What will be the impact of a rise in input price on the supply curve of a commodity?

Ans. The supply curve will shift in the leftward direction because the seller will offer lesser quantity at a given price. (or, the supply curve will shift in the upward direction because the seller will then charge higher price for supplying the same quantity)

3. What is the implication of a vertical supply curve?

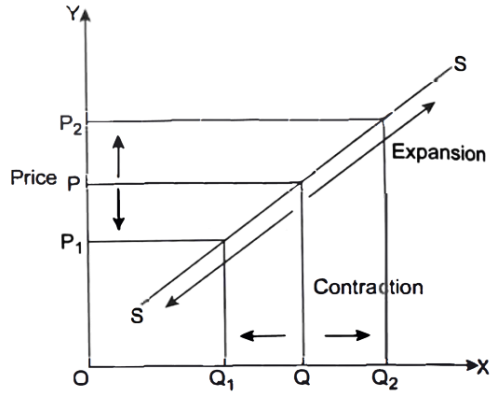
Ans. It implies completely inelastic supply, L.e., quantity supplied remains unchanged even when there is a change in its own price.

4. What do you mean by infinitely elastic supply?

Ans. If there is an infinite change (or a huge change) in quantity supplied in response to a very small change in its price, it is called as infinitely elastic (or completely elastic) supply.

5. What is the implication of movement along any supply curve?

Ans. A movement from one price-quantity combination on a supply curve to another combination on the same curve implies a change in quantity supplied due to a change in the price of the commodity (see graph).



6. Can the supply curve be negatively sloped?

Ans. Generally the supply curve is positively sloped. However, in case of a decreasing-cost industry, it becomes possible tosupply more output at a lower average cost and hence, at a lower price.

So, the supply curve of such a decreasing-cost industry (in a perfectly competitive market) will be downward sloping.

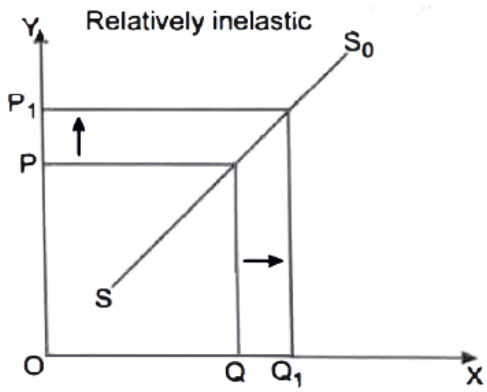
7. Given the price of a commodity, if there is an improvement in production technology, what will be its impact on the supply curve?

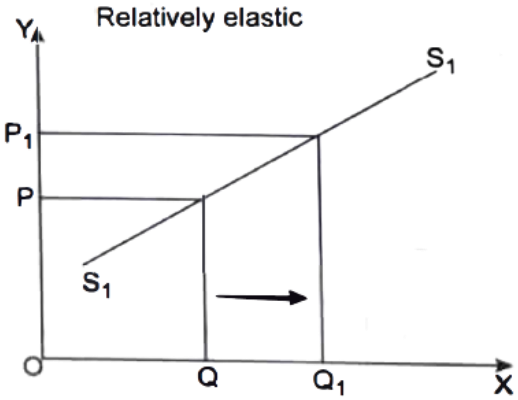
Ans. Given the price of a commodity, an improvement in production technology implies that the producer can now supply more of that commodity at that given price. So, quantity supplied will rise at each price level. This leads to a rightward shift of the supply curve.

8. Draw a relatively elastic and a relatively inelastic supply curve of a commodity.

Ans. Here, the supply curve SS, indicates a relatively inelastic supply. In this case, rate of change in quantity supplied is less than the rate of change in its price. We know that Es <1.

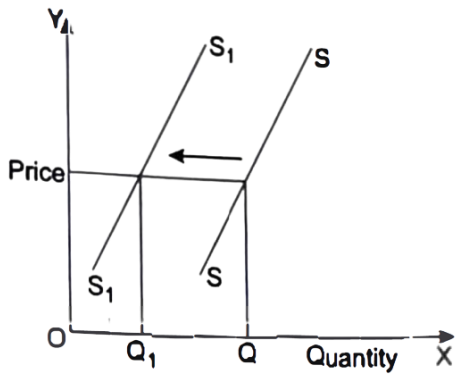
However, the supply curve SS, indicates a relatively elastic supply curve ie., Es>1.0





9. Because of cyclone in a coastal area, the sea water covers a lot of rice fields. This reduces the productivity of land. How will it affect the supply curve of rice of that region?

Ans. Supply curve for rice will shift to the left.



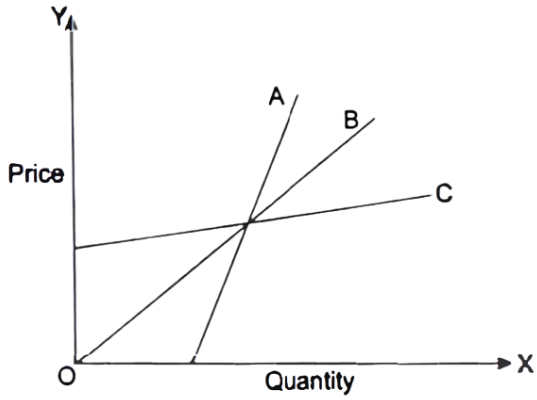
10. If two supply curves intersect each other, which one will have higher elasticity of supply?
Ans. Flatter supply curve will be more elastic than steeper one at the point of intersection.

11. What is the price elasticity of supply of a curve passing through the origin making an angle 60° with OX-axis?
Ans. Elasticity of supply in this case will be unitary elastic, E, = 1.

12. A straight line supply curve cutting Y-axis, what will be the elasticity of supply?
Ans. A supply curve cutting Y-axis will have elasticity more than one, E, >1.

13. A straight line supply curve cutting X-axis, what will be the elasticity of supply?
Ans. A supply curve cutting X-axis will have elasticity less than one, E, <1.

14. The diagram below shows three supply curves of three commodities. Rank their price elasticities.



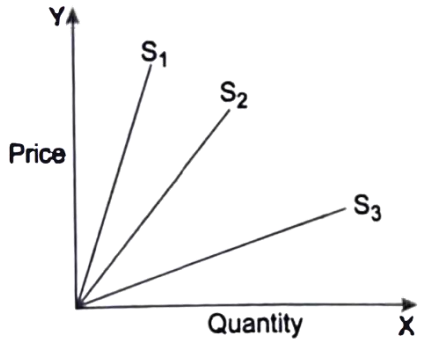
Ans. (i) Supply curve A is less than unit elastic as it cuts X-axis.
(ii) Supply curve B is unitary elastic as it starts from the origin.
(iii) Supply curve C is more than unitary elastic as it starts from Y-axis.

WORKSHEET FOR SELF EVALUATION

1. Derive market supply schedule and curve from the following hypothetical individual supply schedules of two firms A and B: **(2 Marks)**

Price	S_A	S_B	Market Supply
2	10	7	
3	12	8	
4	14	9	

2. The diagram below shows the supply curve of three commodities. Rank their price elasticities: **(2 Marks)**



3. **(4 Marks)**

Price	Firm A	Firm B	Firm C	Market Supply
1	-	-	45	100
2	37	37	50	-
3	40	40	55	135
4	44	44	-	154
5	48	60	65	-

(i) Complete the above table.

(ii) Plot the supply curve of each firm and the market supply curve in a single diagram. What relationship do you observe between the individual supply curve and the market supply curve?

4. If the Government gives production subsidy to any producer, what would be its impact on the supply curve? **(2 Marks)**

5. Why a normal supply curve is called a ceteris paribus supply curve? **(2 Marks)**

6. If a farmer grows rice and wheat, how will an increase in the price of wheat affect the supply curve of rice? **(2 Marks)**

7. Price elasticity of a goods is 0.8. Is the supply elastic or inelastic and why? **(2 Marks)**

8. Differentiate between intended and actual supply. **(2 Marks)**

9. When the market price of cheese rises from 20 per kg to 21 per kg, a producer expands his supply from 3000 kg to 3500 kg. Calculate his price elasticity of demand. **(2 Marks)**

10. Are the following statement true or false? Give reason for your answer. **(3 Marks)**

(i) An increase in input price shifts the supply curve to the right.

(ii) Any reduction in excise tax rate shifts the supply curve to the right.

(iii) An upward movement along the same supply curve indicates increase in supply.